

FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY (FISAT)
DEPARTMENT OF COMPUTER APPLICATIONS
MAIN PROJECT
SCRUM BOOK

Name of the Student: SHAFNA SHAKEER S M

Roll No:41

Batch: B

Email ID: shafnasm20@gmail.com

Name of the Guide: Dr. RAKHI VENUGOPAL

Name of the Scrum Master: Ms. JOICE T

Project Title: FISAT GATEWAY MANAGEMENT SYSTEM

GitHub ID: <https://github.com/shafnashakeersm/Main-Project>

Sprint Release #: 2**Date: 12/03/2025****Description of Work:**

The **FISAT Gateway Management System** is a web-based application designed to digitize and streamline the manual processes involved in managing visitor records, hostel student movements, and staff gate transactions within FISAT College. Built using the MERN stack (MongoDB, Express.js, React.js, Node.js), the system ensures efficient tracking, secure record-keeping, and seamless communication among stakeholders, including wardens, watchmen, parents, and administrators. The platform allows wardens to add and manage student details, including personal information and leave requests, which require parental approval via API-based notifications before students can exit the campus. Once a parent approves, the leave request is forwarded to watchman for final clearance. Additionally, the system enables students traveling by train to upload their ticket details along with their expected return time. The watchman interface provides a real-time view of approved student movements and visitor logs, ensuring that only authorized individuals enter or leave the premises. The admin dashboard consolidates daily reports, including visitor logs, student leave records, and watchman reports, providing complete visibility over campus movements. By implementing role-based access control, the system ensures that wardens, watchmen, and administrators have functionalities tailored to their responsibilities. The integration of automated notifications and approvals enhances transparency, reduces manual errors, and improves overall security within the institution.

Remarks:**Name and Signature of the Guide****Name and Signature of the Scrum Master**

Date: 12/03/2025

PRODUCT BACKLOG
>> User Authentication & Role Management - Implemented a secure login system for different user roles, including Admin, Wardens, and Watchmen. - Ensured role-based access control so that each user only interacts with the necessary functionalities. >> Student Management Module - Wardens can add and manage student records, including student ID, name, batch, class, and parental contact information. - Allows fetching of student details by ID for quick access. >> Student Leave & Approval Module - When a student requests leave, the system notifies parents via API-based messages. - Parents can approve or reject the request directly through a mobile notification. - Once approved, the warden forwards the request to the watchman. - If students are traveling by train, they can upload ticket details along with expected return time. >> Watchman Dashboard & Visitor Management - The watchman receives a real-time list of approved students who are allowed to exit. - Stores visitor details, including name, purpose of visit, entry and exit time for security tracking. >> Notification & Messaging System - Automated API-based notifications are sent to parents when students leave the hostel. - Watchmen receive updates regarding student movements. - Ensures all approvals and actions are logged for future reference. >> Admin Dashboard & Report Generation - The admin can view reports on daily visitor logs, student leave details, and watchman reports. - Provides an overview of student movements, ensuring efficient monitoring and record-keeping.

Name and Signature of the Guide

Name and Signature of the Scrum Master

Date: 12/03/2025

DATABASE & UI DESIGN**DATABASE****>> Student Details**

Field Name	Data Type	Description
_id	ObjectId	Unique ID (Auto-generated by MongoDB)
studentID	String	Unique Student ID
name	String	Student's Full Name
batch	String	Student's Batch (e.g., 2023)
class	String	Class/Department (e.g., CSE, ECE)
parentPhone	String	Parent's Contact Number
createdAt	Date	Record Creation Timestamp

>> Student Leave Record

Field Name	Data Type	Description
_id	ObjectId	Unique ID (Auto-generated)
studentID	String	Student's Unique ID (Reference to <code>students</code> collection)
studentName	String	Student's Name
batch	String	Student's Batch
class	String	Student's Class
parentApproval	Boolean	Whether Parent Approved (true/false)
approvalStatus	String	Status: Pending, Approved, or Rejected
departureTime	Date	Time Student Leaves the Campus
returnTime	Date	Expected Return Time
ticketUploaded	Boolean	If the Train Ticket is Uploaded
createdAt	Date	Record Creation Timestamp

>> Watchman Report

Field Name	Data Type	Description
<code>_id</code>	ObjectId	Unique ID
<code>date</code>	Date	Report Date
<code>studentMovements</code>	Array of Objects	List of students who left/returned
Inside <code>studentMovements</code> Array:		
<code>studentID</code>	String	Student ID (Reference to <code>students</code> collection)
<code>name</code>	String	Student's Name
<code>departureTime</code>	Date	Time of Exit
<code>returnTime</code>	Date	Time of Return

>> Login and Role Management

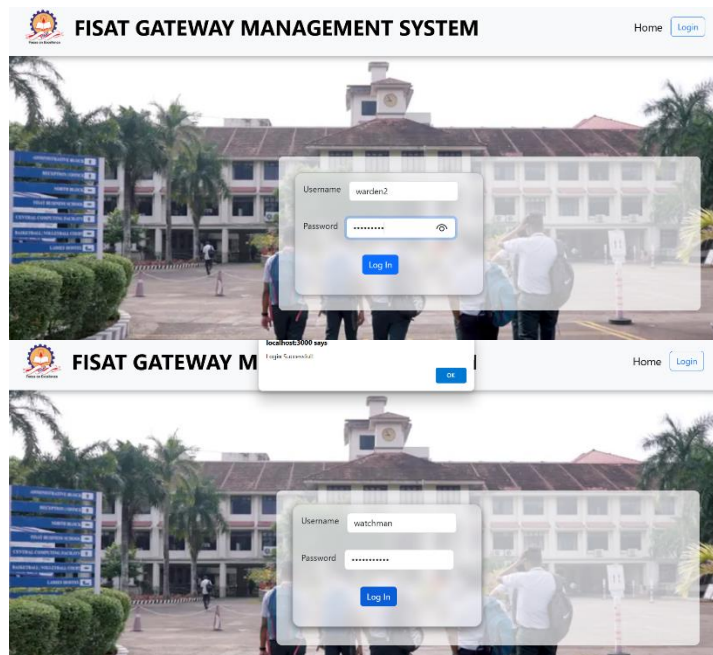
Field Name	Data Type	Description
<code>_id</code>	ObjectId	Unique ID
<code>username</code>	String	User's Username (Unique)
<code>password</code>	String	Encrypted Password (Hashed)
<code>role</code>	String	Role: Admin, Warden, or Watchman
<code>createdAt</code>	Date	Account Creation Date

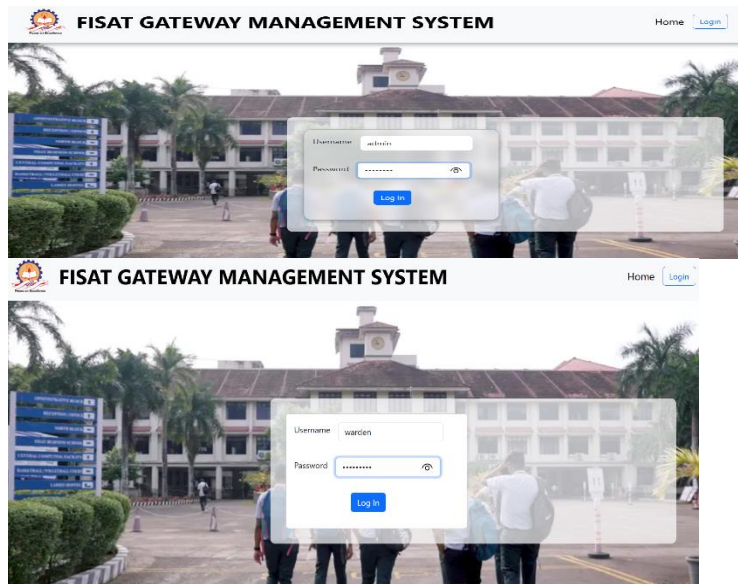
UI DESIGN

Home Page

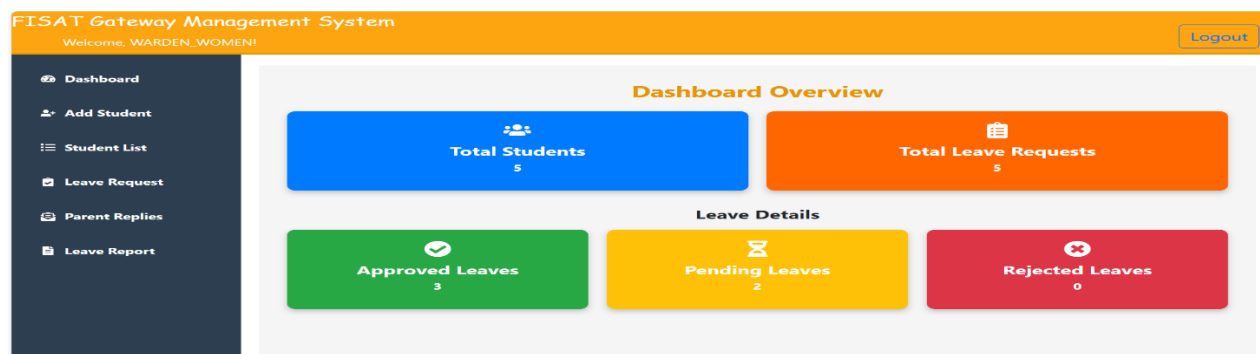
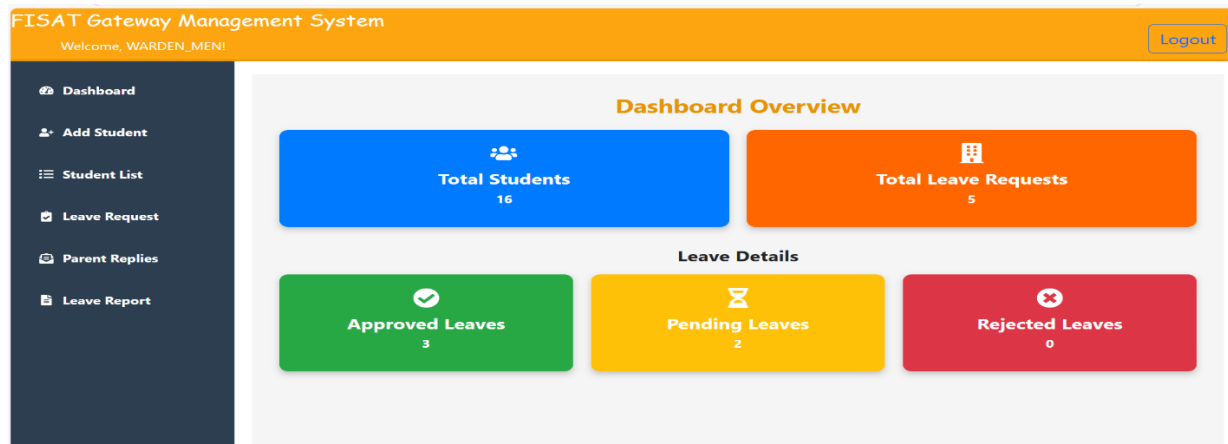


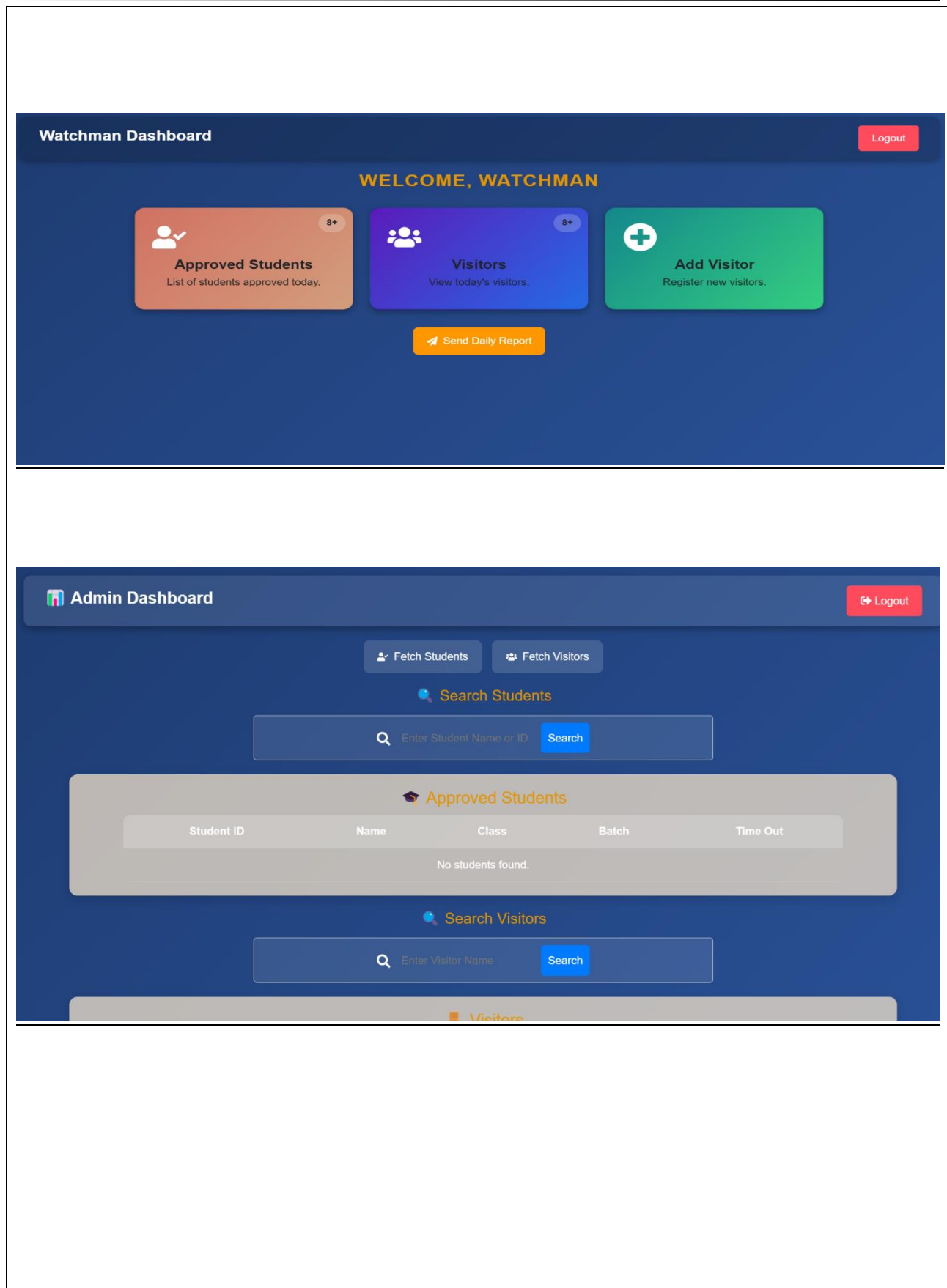
User Management





Dashboard





AddStudent

FISAT Gateway Management System

Welcome, WARDEN_MEN!

Logout

Dashboard

Add Student

Student List

Leave Request

Parent Replies

Leave Report

Add Student

Name:

Nihal Navas

Student ID:

STUD09077

Class:

S3 Btech

Batch:

A

Hostel:

Men's Hostel

FISAT Gateway Management System

Welcome, WARDEN_WOMEN!

Logout

Dashboard

Add Student

Student List

Leave Request

Parent Replies

Leave Report

Add Student

Name:

Asiya M

Student ID:

STUD65432

Class:

S2 MBA

Batch:

B

Hostel:

Women's Hostel

Parent's Phone No:

Student List

FISAT Gateway Management System

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Leave Report

Student List

Search by Name or ID

Name	Student ID	Class	Batch	Hostel	Parent Phone	Parent Email	Actions	
Mahesh J K	STU87643	CSE S3	B	Men's Hostel	754345666	karthi@gmail.com	Edit	Delete
James Antony	STU89065	Btech S2	B	Men's Hostel	5432678654	antony@gmail.com	Edit	Delete
Raahan Muhammod	STU54376	Btech S3	A	Men's Hostel	7543237657	jaafar@gmail.com	Edit	Delete
Ajay Dev	STU546532	Btech S4	A	Men's Hostel	7544543213	dev12@gmail.com	Edit	Delete
Shyam Menon	STU546765	MBA S3	B	Men's Hostel	6457834567	menonjk@gmail.com	Edit	Delete
Noyel S	STU546875	MBA S3	B	Men's Hostel	7543287654	somnath6@gmail.com	Edit	Delete
Sharath John	STUD76549	MBA S4	B	Men's Hostel	7646899987	john@gmail.com	Edit	Delete
Shahas	STUD78658	MCA S4	B	Men's Hostel	7654378965	sageer@gmail.com	Edit	Delete
John James	STUD98567	MBA S4	A	Men's Hostel	5432654765	james@gmail.com	Edit	Delete
Manu Ajay	STUD53446	MCA S4	A	Men's Hostel	77654327654	ajay123@gmail.com	Edit	Delete
Ahaan Mathews	STUD09564	S3 Btech	B	Men's Hostel	6544654321	mathews78@gmail.com	Edit	Delete
Aatham John	STUD54326	S3 Btech	A	Men's Hostel	7643256776	john@gmail.com	Edit	Delete

FISAT Gateway Management System

Welcome, WARDEN_WOMEN!

Logout

Dashboard

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Student List

Search by Name or ID

Name	Student ID	Class	Batch	Hostel	Parent Phone	Parent Email	Actions	
Manju Joyson	STUD09876	S1 MCA	A	Women's Hostel	6564533323	shafnasm20@gmail.com	Edit	Delete
Malavigha M.A	STUD07668	S1 MCA	A	Women's Hostel	7656453433	shefism20@gmail.com	Edit	Delete
Aamina Mujeeb	STUD90765	S3 Btech	B	Women's Hostel	6756454534	mujeeb@gmail.com	Edit	Delete
Divya Krishnan	STUD09755	S3 Btech	B	Women's Hostel	5644356346	krishna@gmail.com	Edit	Delete
Daya P	STUD09765	S1 Btech	A	Women's Hostel	6754654543	shefism20@gmail.com	Edit	Delete
Asiya M	STUD65432	S2 MBA	B	Women's Hostel	6543876543	shafnasm20@gmail.com	Edit	Delete

Leave Request

[Dashboard](#)[Add Student](#)[Student List](#)[Leave Request](#)[Parent Replies](#)[Leave Report](#)

localhost:3000 says
Leave request submitted!

OK

Name: Nihal Navas**Class:** S3 Btech**Batch:** A

Leave Date:

12-03-2025

Time In:

01:31

PM

Time Out:

04:33

PM

Upload Train Ticket:

Choose File

Screensho...51211.png

STUD09077

Fetch Student

Submit Leave Request

Submit Leave Request

Name: Asiya M**Class:** S2 MBA**Batch:** B

Leave Date:

12-03-2025

Time In:

10:37

AM

Time Out:

05:39

PM

Upload Train Ticket:

Choose File

Screensho...51211.png

Screenshot 2025-02-27 151211.png

STUD65432

Fetch Student

Submit Leave Request

Notification Message

Leave Request for Nihal Navas Inbox x



w58805158@gmail.com via sendgrid.net
to me ▾

5:29 AM (3 minutes ago) ☆ 😊 ↩ ⋮

Your child, Nihal Navas, has requested leave on 2025-03-12.

StudentId:STUD09077

Class: S3 Btech

Batch: A

Time Out: 04:33 PM

Time In: 01:31 PM

Please reply with "Approved" or "Rejected" by clicking the link:

https://u50830111.ct.sendgrid.net/ls/click?upn=u001_wO5n5r6G-2FUeHCRjEXZkj7i57AX9IRJG4VXazInE4K8e3upCEI2MPPUgkHkGwhissgSGW_R9xFqYDz5m4AXVGt0hvQtlbQlIX22KoT-2BU-2Fp1OH3gIK2sLEVmXrwcU-2FSU1CMZWX4hDFDQuYrILZOhaRLu6NE7djdC77dt3nS6LLqyta0hqDwLVbD-2BNL-2FaC3qhMSF9vnmPsGlsgVu2uGxTu82W92GGZeqoZAzaPZBoVJs-2FmeBb2V5Bs7RZ4Ejmf5flb8PfbfHU2exmwpZYhhf02HTLjkQ-3D-3D

Leave Request for Asiya M Inbox x



w58805158@gmail.com via sendgrid.net
to me ▾

5:37 AM (0 minutes ago) ☆ 😊 ↩ ⋮

Your child, Asiya M, has requested leave on 2025-03-12.

StudentId:STUD65432

Class: S2 MBA

Batch: B

Time Out: 05:39 PM

Time In: 10:37 AM

Please reply with "Approved" or "Rejected" by clicking the link:

https://u50830111.ct.sendgrid.net/ls/click?upn=u001_wO5n5r6G-2FUeHCRjEXZkj7i57AX9IRJG4VXazInE4K8e3upCEI2MPPUgkHkGwhissAB0J_R9xFqYDz5m4AXVGt0hvQlFkqwKqv7HHEPl-2ByoxbPgUpBgwuG8c1PP69JVkeVNwoocM1rhuGGWLFiBECSou5aJQ42Ef1hFBLuA-2F8xe7LsZem8BLGoxMcsQ5UWd3-2FFmQM-2FOaO5mBpzG2SEFco-2FWFZB-2FEFf6Dku86gmbCL-2FKyZV-2Fgh-2FukokEIEBjDoZ-2FFQ9bcUWO-2FQ1v-2Fbcvm4P29n5mHrqGg-3D-3D

Parent Reply

Parent Replies

Student ID	Name	Class	Batch	Leave Date	Time Out	Parent Reply	Send to Watchman
STUD43657	Nameer Hussain	S2 MBA	B	2025-03-08	04:13 PM	Approved	<button>Send to Watchman</button>
STUD65457	Abay Krish	S1 MCA	A	2025-03-20	04:28 PM	Approved	<button>Send to Watchman</button>
STUD43657	Nameer Hussain	S2 MBA	B	2025-03-11	05:23 PM	Pending	✗ Not Approved
STUD65457	Abay Krish	S1 MCA	A	2025-03-11	05:39 PM	Pending	✗ Not Approved
STUD80975	Abinav J	S3 MCA	B	2025-03-11	05:43 PM	Approved	<button>Send to Watchman</button>
STUD09077	Nihal Navas	S3 Btech	A	2025-03-12	04:33 PM	Approved	<button>Send to Watchman</button>

FISAT Gateway Management System

Welcome, WARDEN_MENI

Logout

Dashboard

Add Student

Student List

Leave Request

Parent Replies

Leave Report

Parent Replies

Student ID	Name	Class	Batch	Leave Date	Time Out	Parent Reply	Send to Watchman
STUD43657	Nameer Hussain	S2 MBA	B	2025-03-08	04:13 PM	Approved	<button>Send to Watchman</button>
STUD65457	Abay Krish	S1 MCA	A	2025-03-20	04:28 PM	Approved	<button>Send to Watchman</button>
STUD43657	Nameer Hussain	S2 MBA	B	2025-03-11	05:23 PM	Pending	✗ Not Approved
STUD65457	Abay Krish	S1 MCA	A	2025-03-11	05:39 PM	Pending	✗ Not Approved
STUD80975	Abinav J	S3 MCA	B	2025-03-11	05:43 PM	Approved	<button>Send to Watchman</button>
STUD09077	Nihal Navas	S3 Btech	A	2025-03-12	04:33 PM	Approved	<button>Send to Watchman</button>

Leave Report

FISAT Gateway Management System

Welcome, WARDEN_MEN!

Logout

Dashboard

Add Student

Student List

Leave Request

Parent Replies

Leave Report

Leave Reports

From: 28-02-2025 To: 20-03-2025 [Generate Report](#)

Student ID	Name	Class	Batch	Leave Date	Time Out	Parent Reply
STUD43657	Nameer Hussain	S2 MBA	B	2025-03-08	04:13 PM	Approved
STUD65457	Abay Krish	S1 MCA	A	2025-03-20	04:28 PM	Approved
STUD43657	Nameer Hussain	S2 MBA	B	2025-03-11	05:23 PM	Pending
STUD65457	Abay Krish	S1 MCA	A	2025-03-11	05:39 PM	Pending
STUD80975	Abinav J	S3 MCA	B	2025-03-11	05:43 PM	Approved
STUD09077	Nihal Navas	S3 Btech	A	2025-03-12	04:33 PM	Approved

FISAT Gateway Management System

Welcome, WARDEN_WOMEN!

Logout

Dashboard

Add Student

Student List

Leave Request

Parent Replies

Leave Report

Leave Reports

From: 01-03-2025 To: 28-03-2025 [Generate Report](#)

Student ID	Name	Class	Batch	Leave Date	Time Out	Parent Reply
STUD09876	Manju Joyson	S1 MCA	A	2025-03-12	04:26 PM	Approved
STUD09876	Manju Joyson	S1 MCA	A	2025-03-05	04:30 PM	Pending
STUD09876	Manju Joyson	S1 MCA	A	2025-03-12	05:29 PM	Pending
STUD07668	Malavigha M.A	S1 MCA	A	2025-03-05	05:13 PM	Approved
STUD09765	Daya P	S1 Btech	A	2025-03-11	03:52 PM	Approved
STUD65432	Asiya M	S2 MBA	B	2025-03-12	05:39 PM	Approved

Watchman page

Approved Students

← Back

Search by Name or Student ID ..

Name	Student ID	Leave Date	Time Out	Warden Approval
Manju Joyson	STUD09876	2025-03-12	04:26 PM	Approved
Sameer Hussain	STUD43657	2025-03-08	04:13 PM	Approved
Abay Krish	STUD65457	2025-03-20	04:28 PM	Approved
Malavigha M.A	STUD07668	2025-03-05	05:13 PM	Approved
Abinav J	STUD80975	2025-03-11	05:43 PM	Approved
Daya P	STUD09765	2025-03-11	03:52 PM	Approved
Nihal Navas	STUD09077	2025-03-12	04:33 PM	Approved
Aasiya M	STUD65432	2025-03-12	05:39 PM	Approved

Add Visitor

Ram P

6543567788

KL764325

Gas Supply

Entry Time:

01:50 PM

Exit Time (Optional):

01:52 PM

Date:

12-03-2025

Add Visitor

Back to Dashboard

VISITOR LIST

Back

Search by Name or Mobile Number ..

Name	Mobile	Vehicle	Purpose	Entry	Exit	Date
John Jaison	6543278543	KL876543	Carleen	10:21	11:26	2025-03-29
Mathew Johnson	5435632543	KL053326	Delivery	10:08	10:14	2025-03-21
Martin Dav	6566543543	KL639534	Gas Supply	11:59	12:06	2025-03-18
Mathav J	5456463343	KL765433	Carleen	02:21	03:22	2025-03-13
Nikitha Sanjay	7656454343	KL765433	Meeting	10:17	03:15	2025-03-12
Ram P	6543567788	KL764325	Gas Supply	01:50	01:52	2025-03-12
Manav A R	6579453423	KL85432	Food Delivery	01:19	01:21	2025-03-11
Abay Martin	6543466788	KL765433	Meet Principal	03:25	05:00	2025-03-09

Admin Page

 Admin Dashboard
Logout

 Fetch Students
 Fetch Visitors

 Search Students

Search

 Approved Students

Student ID	Name	Class	Batch	Time Out
STUD09876	Manju Joyson	S1 MCA	A	04:26 PM
STUD43657	Nameer Hussain	S2 MBA	B	04:13 PM
STUD65457	Abay Krish	S1 MCA	A	04:28 PM
STUD07668	Malavigha M.A	S1 MCA	A	05:13 PM
STUD80975	Abinav J	S3 MCA	B	05:43 PM

 Search Visitors

Search

 Visitors

Name	Purpose	Vehicle No	Time In	Time Out
John Jaison	Canteen	KL876543	10:21	11:26
Mathew Johnson	Delivery	KL653326	10:08	10:14
Martin Dev	Gas Supply	KL639534	11:59	12:06
Mathav J	Canteen	KL765433	02:21	03:22
Nikitha Sanjay	Meeting	KL765433	10:17	03:15
Ram P	Gas Supply	KL764325	01:50	01:52

Name and Signature of the Guide:

Name and Signature of the Scrum Master

Date: 12/03/2025

TESTING AND VALIDATION

Name and Signature of the Guide:

Name and Signature of the Scrum Master

Date: 12/03/2025

DETAILS OF VERSIONS
1. Version Numbering - Follow a structured format like v1.0, v1.1, v2.0, or Semantic Versioning (e.g., Major.Minor.Patch – 1.0.0). Major – Significant changes or new features. Minor – Small feature updates or enhancements. Patch – Bug fixes or security updates. 2. Release Date - The date when the version was deployed or made available. 3. Features & Changes - List new features, enhancements, and modifications made in each version. - Mention improvements in UI, performance, database changes, or added functionalities. 4. Bug Fixes - Document resolved issues and defects in each release. - Include descriptions of the problems fixed and their impact on the system. 5. Known Issues (If Any) - Any existing issues or limitations that are yet to be fixed in future versions. 6. Compatibility & Dependencies - Mention if the version has specific requirements like database migrations, API changes, or compatibility issues with previous versions.

Name and Signature of the Guide

Name and Signature of the Scrum Master